

IMPACT OF SOCIAL MEDIA USAGE ON ACADEMIC PERFORMANCE OF COLLEGE STUDENTS

Team Name : Code Crushers

Team Members

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Project Title

Impact of Social Media Usage on Academic Performance of College Students

GitHub Repository URL

https://github.com/ravi-473/Smart_Student_Analytics

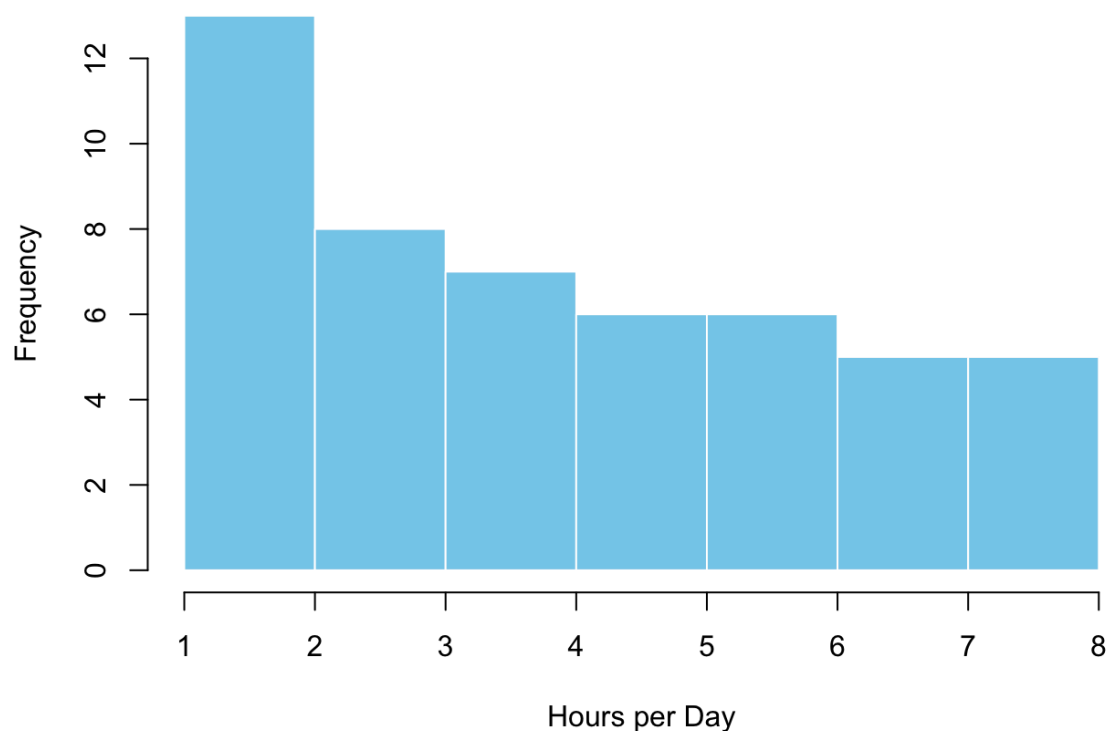
Dataset URL:

https://github.com/ravi-473/Smart_Student_Analytics/blob/main/student_data.csv

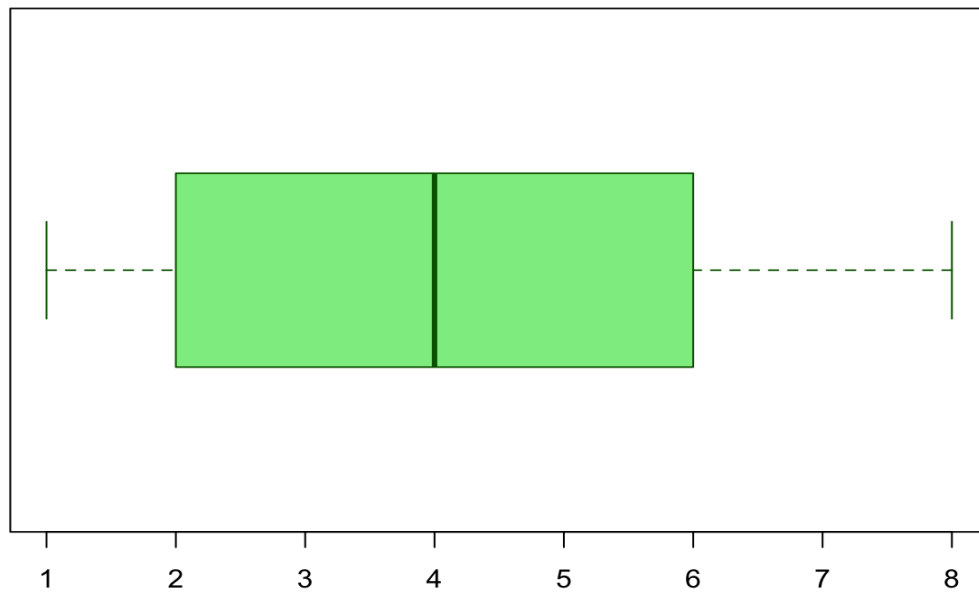
Steps to Run the Project

1. Install R and RStudio.
 2. Download the project files from GitHub.
 3. Open the project folder in RStudio.
 4. Ensure that the file **student_data.csv** is present in the project directory.
 5. Open **project.r**.
 6. Run the script.
 7. The program automatically generates:
 - Dataset Summary
 - Histogram
 - Boxplot
 - Scatter Plot
 - Correlation Analysis
 - Best Student Information
 - Worst Student Information
 - Average Marks Table
-

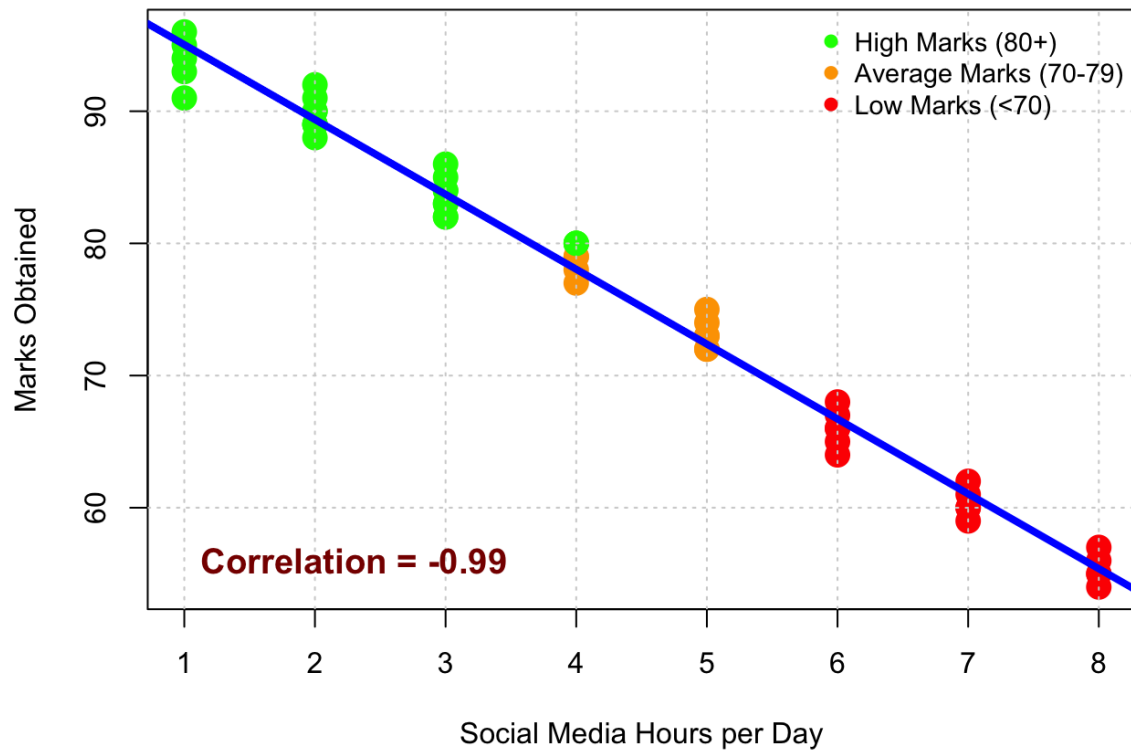
Distribution of Social Media Usage



Boxplot of Social Media Usage



Social Media Usage vs Academic Performance



Dataset Description

The dataset contains information about 50 college students and includes the following variables:

- Student ID
 - Social Media Hours
 - Study Hours
 - Sleep Hours
 - Attendance
 - Marks
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Project Objective

The objective of this project is to analyze how social media usage affects the academic performance of college students. The project uses R programming to perform data analysis, visualization, and statistical correlation analysis.

The study aims to identify whether increased social media usage has any significant impact on students' marks.

Software and Tools Used

- R Programming Language
 - RStudio
 - GitHub
 - CSV Dataset
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Data Analysis Performed

1. Dataset Summary

The summary statistics provide information about the minimum, maximum, mean, median, and quartile values of all variables.

Key Statistics

- Mean Social Media Usage: 4.28 hours/day
 - Median Social Media Usage: 4 hours/day
 - Maximum Social Media Usage: 8 hours/day
 - Minimum Social Media Usage: 1 hour/day
 - Mean Marks: 76.46
 - Median Marks: 79
 - Maximum Marks: 96
 - Minimum Marks: 54
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2. Histogram Analysis

A histogram was generated to visualize the distribution of social media usage among students.

Observation

Most students spend between 2 and 6 hours per day on social media.

3. Boxplot Analysis

A boxplot was generated to understand the spread of social media usage.

Observation

The data is reasonably distributed and does not contain any significant outliers.

4. Scatter Plot Analysis

A scatter plot was generated to study the relationship between Social Media Hours and Marks.

A regression line was also added to better understand the trend.

Observation

The scatter plot clearly shows that marks decrease as social media usage increases.

5. Correlation Analysis

The correlation coefficient between Social Media Hours and Marks was calculated using the `cor()` function in R.

Result

Correlation = **-0.99**

Interpretation

A correlation value close to **-1** indicates a very strong negative relationship between social media usage and academic performance.

This means students who spend more time on social media generally score lower marks.

Best Student Analysis

The student with the highest marks was identified using R.

Observation

The highest-performing student achieved 96 marks while maintaining relatively lower social media usage.

Worst Student Analysis

The student with the lowest marks was identified using R.

Observation

The lowest-performing student scored 54 marks and had higher social media usage compared to top-performing students.

Average Marks by Social Media Usage

The average marks were calculated for each level of social media usage using the **aggregate()** function.

Observation

Students spending fewer hours on social media achieved significantly higher average marks compared to students spending more hours online.

Findings

1. Social media usage and marks have a strong negative relationship.
 2. Students spending fewer hours on social media generally score higher marks.
 3. Academic performance decreases as social media usage increases.
 4. The correlation coefficient of -0.99 confirms a very strong inverse relationship.
 5. Data visualization techniques helped identify important patterns and trends in student behavior.
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Conclusion

This project successfully analyzed the relationship between social media usage and academic performance using R programming.

The analysis revealed a very strong negative correlation between social media usage and marks. Students who spend excessive time on social media tend to achieve lower academic scores, while students with lower social media usage generally perform better academically.

The project demonstrates how statistical analysis and data visualization can be used to discover meaningful insights from educational data and support data-driven decision making.
